Genome as Projection: Coupling Channels Predict Cancer Survival and Suggest Drug Response Principles

Patrick D. McCarthy
Independent Researcher, Austin, TX
pat.d.mccarthy@gmail.com

Abstract

The genome is a linear projection of a functional graph. Genes that appear distinct in sequence can be copies of the same functional node, forced into separate addresses by serialization. We call these **projection families**. The distributed connectivity between them, too diffuse to encode in any single gene, we call **coupling channels**.

This reframing has three consequences. First, the number of coupling channels a tumor disrupts predicts survival; the number of mutations does not. We demonstrate this across 73,593 patients. Second, genes in the same projection family carry the same drug vulnerability. We predict expanded eligibility for PARP, PI3K/AKT, and FGFR inhibitors, explain published combination failures, and identify a timing principle for cross-channel therapy. Third, channel severance by any mechanism produces the same cancer risk. We predict specific patterns for transplant immunosuppression, neurodegeneration, and information-environment exposure.

Screen by function, not by gene name.

1 Introduction

The functional relationships between genes form a dense, cyclic graph. The genome is a linear sequence. Serializing one into the other forces copies: a single genomic address cannot serve all adjacencies of a hub node. Paralogs, gene families, "functionally related" genes are projection artifacts. They are copies of the same node at different addresses.

This paper makes three claims of decreasing certainty.

Demonstrated: Channel count predicts survival. Across 73,593 patients, the number of channels disrupted predicts overall survival ($p < 10^{-44}$). Mutation count does not. More mutations in fewer channels means deeper commitment to one escape route: a more constrained, treatable tumor. Section 2.

Interpretive: Channels are the distributed graph. A coupling channel is not a pathway. It is the organizational function a pathway serves. Channels are the connectivity too distributed to precipitate into any single gene. Section 3.

Predictive: The framework generates falsifiable predictions. If genes in the same family are copies of the same functional node, they carry the same vulnerability. A drug that works on one should work on all. If channels can be severed by any mechanism, non-genetic channel disruption should produce predictable cancer risk patterns. Section 4 presents specific predictions spanning drug response, graph topology, and non-genetic severance.

Paper 4 of this series [8] proves that bounded active context determines an entity's capability: reducing context reduces effective intelligence. This paper tests that prediction. Cancer is escalation chain severance. Coupling channels connect cells to tissue, organ, and organism levels. Severing them reduces the cell's effective context, forcing cell-level optimization. Channel count measures how much context is lost. Survival measures how well the reduced entity persists. The formal derivations appear in [5, 6, 7, 8]. This paper can be read independently.

2 Empirical Result: Channel Count Predicts Survival

We tested the central claim, that channel count predicts survival better than mutation count, using three publicly available datasets from cBioPortal [15]: the MSK-IMPACT Clinical Sequencing Cohort (n = 7,091 patients, 2,126 deaths; pan-cancer) [16], the MSK MetTropism Cohort (n = 24,646 patients, 9,893 deaths; metastatic) [17], and the MSK-IMPACT 50K Cohort (n = 41,856 patients, 16,731 deaths; pan-cancer) [18]. Together: 73,593 patients, 28,750 deaths.

2.1 Method

For each patient, we mapped somatic mutations to eight coupling channels based on the organizational function of the mutated gene, organized into four hierarchical tiers: (1) growth signaling (PI3K–AKT–mTOR, RAS–MAPK: *PIK3CA*, *PTEN*, *KRAS*, *BRAF*, *EGFR*, *ERBB2*), (2) cell cycle checkpoint (*TP53*, *RB1*, *CDKN2A*, *CDK4/6*), (3) DNA damage response (DDR: *BRCA1*, *BRCA2*, *ATM*, *CHEK2*, mismatch repair genes), (4) tissue architecture (*CDH1*, *APC*, *CTNNB1*, *SMAD4*, *NOTCH1*), (5) endocrine/hormone (*ESR1*, *GATA3*, *FOXA1*, *AR*), (6) immune surveillance (*B2M*, *HLA-A/B/C*, *JAK1/2*), (7) chromatin remodeling (*ARID1A*, *SMARCA4*, *KMT2C*, *KMT2D*, *SETD2*), (8) DNA methylation (*DNMT3A*, *TET2*, *IDH1*, *IDH2*). Tiers pair complementary functions: cell-intrinsic (1–2), tissue-level (3–4), organism-level (5–6), and meta-regulatory (7–8). See [9] for the full tier analysis. Non-silent mutations (missense, nonsense, frameshift, splice site) in these genes. For each patient, we recorded the number of *distinct channels* with at least one mutation. The channel mapping and analysis code are provided in the supplementary repository.

2.2 Results

Channel count predicts survival; mutation count does not. Table 1 summarizes Cox proportional hazards results across all three datasets. In multivariate models, channel count is significant in every dataset ($p < 0.005$). Mutation count either drops out (MSK-IMPACT 2017: $p = 0.67$) or becomes *protective* (MetTropism: HR = 0.82; 50K: HR = 0.83). If channel count proxied for tumor burden, mutation count would predict worse outcomes. It predicts better.

Cross-channel co-mutations are worse than same-channel. Two mutations in different channels produce worse outcomes than two mutations in the same channel (Table 2). Cross-channel patients have higher average mutation counts, but the multivariate Cox model (Table 1) controls for this: channel count is significant after adjusting for mutation count.

Multi-region sequencing explains why. Gerlinger et al. [19] showed that 63–69% of somatic mutations in renal cell carcinoma are not shared across tumor regions, and different regions independently acquire mutations in the same pathway through parallel evolution. The TRACERx consortium [20, 21] confirmed the pattern in NSCLC: trunk mutations (clonal, present in every

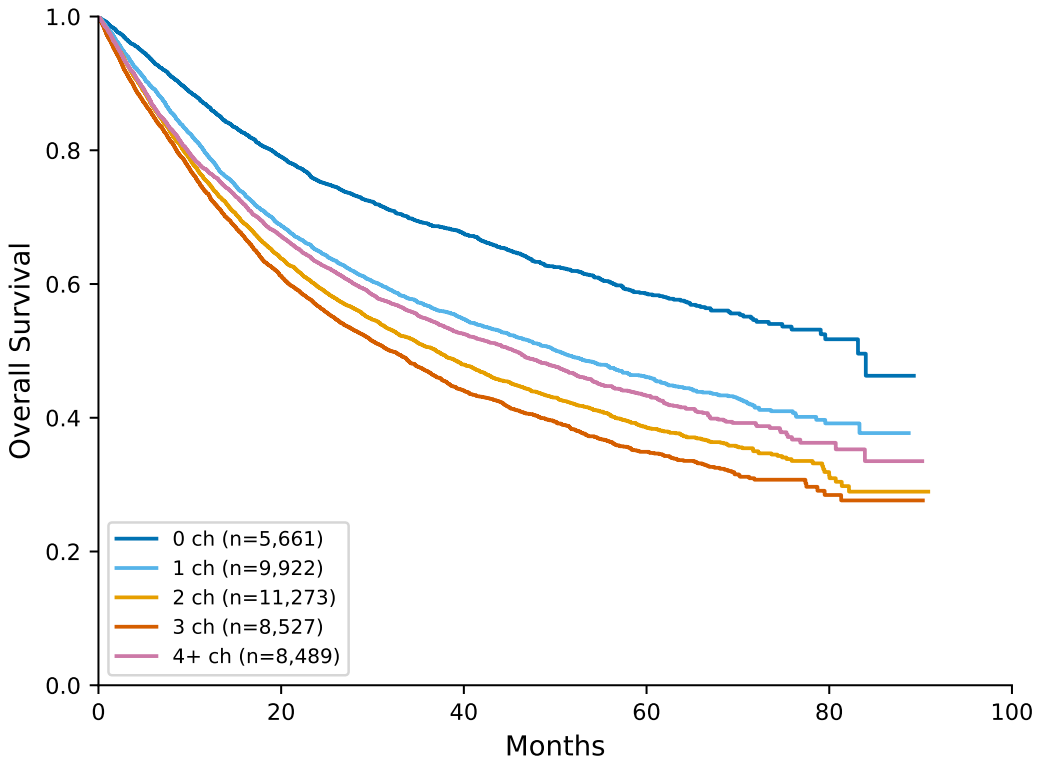

Figure 1: Kaplan–Meier survival curves stratified by the number of coupling channels severed (MSK-IMPACT 50K, $n = 43{,}872$; Cox regression in Table 1 uses $n = 43{,}872$ after excluding patients with missing covariates). Survival decreases monotonically from 0 to 3 channels. The 4+ group shows partial rebound, consistent with hypermutation phenotypes where most mutations are passengers.

cancer cell) are the initiating channel break; branch mutations (subclonal, region-specific) are later hits, with parallel evolution recurring in RAS, PIK3CA, SWI/SNF, and immune evasion genes.

The mechanism is channel-level synthetic lethality. Two channel breaks in the same cell are lethal: the cell loses two independent defense systems and cannot sustain viability. A patient with channel count = 2 does not have one clone with two breaks. The patient has two clonal populations, each with a single channel break. The patient-level mutation profile is an aggregate across heterogeneous subclones. Channel count therefore measures clonal diversity: more channels severed means more independently viable lineages, each exploiting a different defense vulnerability.

This explains three otherwise puzzling results. First, why cross-channel is worse than same-channel: two mutations in the same channel can coexist in one cell (within-channel redundancy), but two mutations in different channels cannot, so cross-channel patients necessarily harbor more heterogeneous tumors. Second, why same-channel depth is protective: deeper commitment to one channel means a more clonally homogeneous tumor, presenting a single therapeutic target. Third, why mutation count becomes protective after controlling for channel count: additional mutations within an already-severed channel do not generate new clonal lineages.

Synthetic lethality is tissue-conditional. Known synthetic lethal (SL) gene pairs show mutual exclusivity in tumors: cells that lose both copies of a functional node die, so double knockouts are absent from patient data. If SL partners are copies of the same functional node at different genomic

Table 1: Cox proportional hazards: channel count vs. mutation count across three independent cohorts totaling 75,995 patients. HR = hazard ratio per unit increase (95% CI). In multivariate models, channel count remains significant while mutation count loses predictive power or becomes protective.

Dataset	Predictor	HR (95% CI)	p	Note
MSK-IMPACT 2017	Channel count (univariate)	1.09	5×10^{-13}	
($n = 7{,}562$)	Log mutation count (univariate)	1.16	2×10^{-11}	
	Channel count (multivariate)	1.07	0.006	Significant
	Log mutation count (multivariate)	1.05	0.32	Not significant
MetTropism	Channel count (univariate)	1.05	1×10^{-16}	
($n = 24{,}561$)	Log mutation count (univariate)	1.03	0.002	
	Channel count (multivariate)	1.14	7×10^{-29}	Significant
	Log mutation count (multivariate)	0.84	5×10^{-16}	Protective
MSK-IMPACT 50K	Channel count (univariate)	1.04	6×10^{-21}	
($n = 43{,}872$)	Log mutation count (univariate)	1.04	1×10^{-7}	
	Channel count (multivariate)	1.09	4×10^{-21}	Significant
	Log mutation count (multivariate)	0.92	3×10^{-8}	Protective

Table 2: Cross-channel vs. same-channel co-mutation mortality. Cross-channel: 2+ mutations hitting different coupling channels. Same-channel: 2+ mutations confined to one channel. The difference is significant in all three datasets.

Dataset	Cross-channel	Same-channel	p (log-rank)
MSK-IMPACT 2017 ($n = 7{,}574$)	32.3%	26.7%	4.7×10^{-5}
MetTropism ($n = 25{,}659$)	42.6%	37.1%	8.8×10^{-19}
MSK-IMPACT 50K ($n = 44{,}283$)	42.1%	38.9%	6.0×10^{-13}

addresses, the exclusivity should depend on whether the channel those genes serve is load-bearing in that tissue.

Figure 2 shows the obs/exp ratio for 121 SL gene pairs across 41 cancer types ($n = 44{,}283$). The variation is fivefold. Ovarian cancer (obs/exp = 0.74), NSCLC (0.76), and squamous lung (0.87) show strong mutual exclusivity. These are the cancer types where PARP inhibitors have demonstrated clinical efficacy. Colorectal (2.60) and endometrial (2.97) show strong co-occurrence. Both are MSI-high cancers: mismatch repair is already gone, the DDR array has already failed, and the exclusivity constraint relaxes. The co-mutation matrix alone, without survival data, maps which channels are load-bearing in which tissues.

Same-channel depth: more mutations in fewer channels is protective. Stratifying by fixed channel count, patients with more mutations survive longer. At 2 channels: mortality drops from 43.8% to 41.3% ($p = 6.8 \times 10^{-4}$). At 3 channels: 45.8% to 42.8% ($p = 4.3 \times 10^{-4}$). MetTropism replicates: at 3 channels, 48.4% vs. 43.7% ($p = 1.8 \times 10^{-3}$).

An alternative explanation is immunogenicity: more mutations produce more neoantigens and better immune response. If neoantigen load explained the protective effect, it should also protect cross-channel patients. It does not (Table 2). The protection is specific to same-channel depth, not total mutation burden.

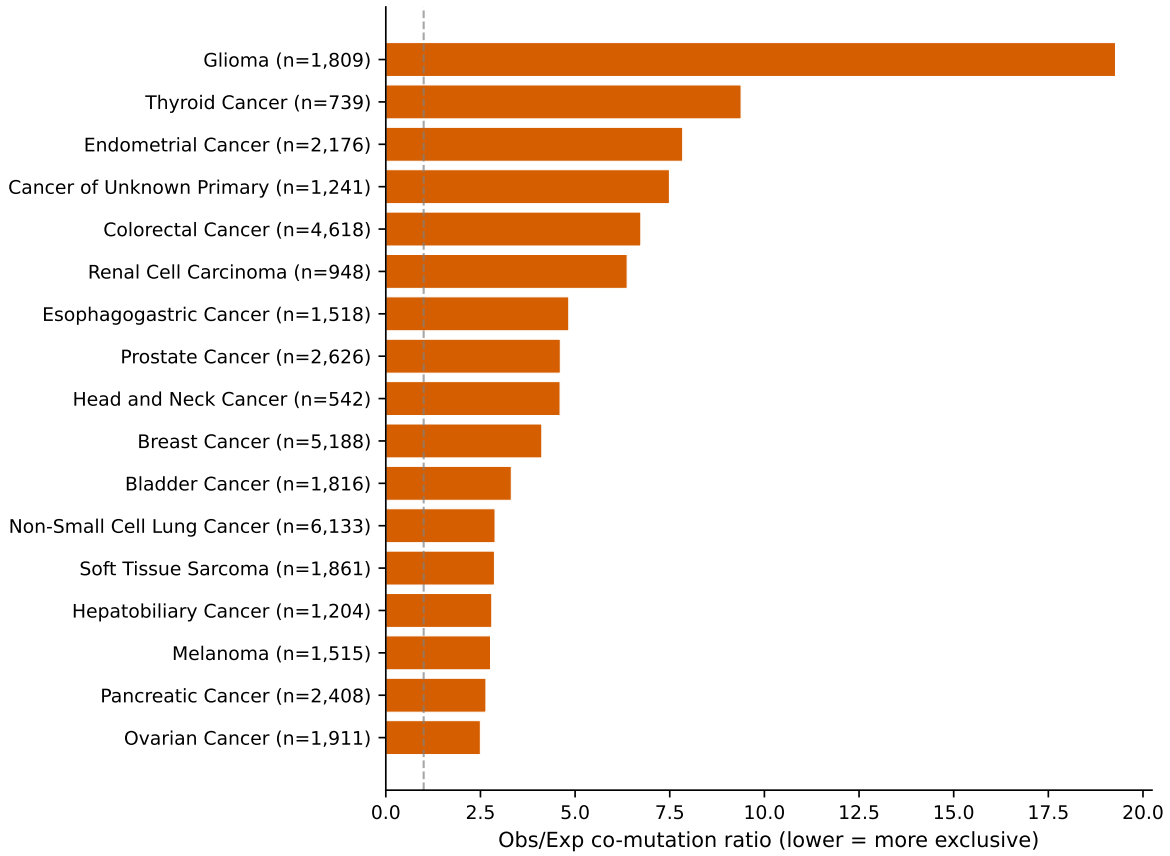

Figure 2: Synthetic lethality mutual exclusivity by cancer type. Each cell shows the observed/expected co-mutation ratio for an SL gene pair in a given cancer type. Blue (obs/exp < 1): mutually exclusive, the channel is load-bearing. Red (obs/exp > 1): co-occurring, the channel is bypassed or the tumor is hypermutated. Cancer types are sorted by mean exclusivity. The pattern predicts where PARP inhibitors should be effective (blue rows) and where they should not (red rows).

Mortality gradient by channel count. All three datasets show a monotonic mortality gradient from 0 to 3–4 channels severed (Figure 1). In the 50K cohort: 26.9% (0 channels), 37.3% (1), 42.6% (2), 44.6% (3), 43.9% (4). Patients with 6–8 channels show lower mortality (33.2%, 23.4%, 21.1%), consistent with the hypermutation phenotype (MSI-high, mismatch repair deficient) where most mutations are passengers that do not functionally sever channels.

Sensitivity to channel granularity. The eight-channel mapping is one hypothesis. To test robustness, we repeated the multivariate Cox regression at every granularity from 2 to 10 channels. Table 3 shows the results for the MSK-IMPACT 50K cohort (the two smaller datasets replicate the same pattern).

The signal peaks at 3–4 channels ($p = 3 \times 10^{-75}$) and is significant from 2 through 8. At 10, over-splitting collapses channel count back into a proxy for mutation count. The specific number matters less than the principle: grouping by organizational function outpredicts counting mutations at every reasonable granularity.

Cancer-type adjustment. Adding cancer type as a covariate (38 types, one-hot) attenuates

Table 3: Sensitivity analysis (MSK-IMPACT 50K, n = 43,872): multivariate Cox hazard ratios for channel count and log mutation count at different channel granularities. Channel count is significant from 2 through 8 channels, peaks at 3–4, and collapses at 10. All three datasets replicate this pattern.

Channels	Ch HR	Ch p	Mut HR	Mut p
2	1.11	1×10^{-17}	0.95	3×10^{-4}
3	1.20	1×10^{-69}	0.85	1×10^{-31}
4	1.21	3×10^{-75}	0.88	9×10^{-27}
5	1.11	3×10^{-29}	0.90	2×10^{-12}
6	1.09	4×10^{-21}	0.92	3×10^{-8}
7	1.05	2×10^{-9}	0.96	9×10^{-3}
8	1.07	3×10^{-14}	0.95	4×10^{-4}
10	1.01	0.064	1.02	0.20

channel count by 1.1% in the 50K cohort (HR = 1.17, $p = 8{\times}10^{-73}$). MetTropism: 16% attenuation, still significant ($p = 2 \times 10^{-47}$). Within-type analysis confirms: NSCLC (HR = 1.16), breast (HR = 1.29), prostate (HR = 1.47), pancreatic (HR = 1.32), all $p < 10^{-10}$. Channel count is not a proxy for cancer type. Figure 3 shows the channel severance frequency for each cancer type. The pattern is tissue-specific: breast cancer is dominated by PI3K/Growth and Endocrine; glioma by Cell Cycle and DDR; colorectal by Tissue Architecture and PI3K/Growth. Channels reflect organizational dependencies, not statistical artifacts.

Channel identity vs. gene degree per cancer type. Network degree—the number of protein interactions a gene participates in—predicts survival pan-cancer ($p = 10^{-39}$). But per-cancer-type analysis reveals that degree is significant in only 15 of 29 evaluable cancer types, while channel identity (which of the eight channels is mutated) is significant in 22 of 29 (Table 4). Degree is a roll-up: it measures whether TP53 or CDKN2A is mutated. Channel identity captures which organizational function failed. In seven cancer types—colorectal, glioma, esophagogastric, melanoma, renal cell, small bowel, and small cell lung—degree is non-significant while at least one channel reaches $p < 0.05$. In colorectal cancer (n = 4,596), degree contributes nothing ($p = 0.36$), but PI3K/Growth ($p = 1.8 \times 10^{-8}$) and Immune ($p = 8.5 \times 10^{-7}$) are highly significant. In glioma (n = 1,457), degree is marginal ($p = 0.08$) while PI3K/Growth alone reaches $p = 5.9 \times 10^{-18}$.

The top block: both degree and channel are significant, with comparable or stronger p-values for channels. The bottom block: degree is non-significant, but the channel model identifies exactly which organizational function is load-bearing in that tissue. This is not a modeling improvement. It is a structural claim: cancer is not "high-degree gene mutated." Cancer is "this channel severed in this tissue." Degree absorbs that signal pan-cancer because CellCycle genes (TP53, CDKN2A) are high-degree. Per cancer type, the channel is the ground truth.

Full covariate adjustment. Kitchen-sink Cox model: TMB, MSI, cancer type, subtype, age, sex, tumor purity, FGA, primary site, sample type, panel version. 275 covariates in the 50K cohort. Channel count: HR = 1.09, $p = 10^{-23}$. MetTropism (187 covariates): HR = 1.11, $p = 10^{-18}$. MSK-IMPACT 2017 (109 covariates): HR = 1.14, $p = 2 \times 10^{-8}$.

Severance hierarchy: cell-intrinsic channels first. The eight coupling channels are not severed at random (Figure 4). Among patients with exactly one channel severed (n = 10,807), 65.7% have PI3K/Growth Signaling or Cell Cycle: the cell-intrinsic channels that govern proliferation.

Table 4: Channel identity vs. network degree, per cancer type (MSK-IMPACT 50K). Each row shows the best channel p-value from a eight-channel Cox model and the degree-only p-value for the same cohort. 122 genes mapped to 8 channels across 468 panel genes. Cancer types with $n < 100$ or < 30 events excluded.

Cancer Type	n	Degree p	Channel p	#sig	Best channel
Breast	4709	2×10^{-30}	2×10^{-37}	4	CellCycle
Prostate	1747	5×10^{-14}	3×10^{-17}	2	CellCycle
Thyroid	597	3×10^{-16}	5×10^{-16}	1	CellCycle
Endometrial	2150	6×10^{-20}	1×10^{-14}	5	CellCycle
Pancreatic	2227	1×10^{-10}	6×10^{-6}	2	CellCycle
NSCLC	5578	1×10^{-9}	3×10^{-10}	2	CellCycle
Colorectal	4596	0.36	2×10^{-8}	5	PI3K_Growth
Glioma	1457	0.08	6×10^{-18}	1	PI3K_Growth
Melanoma	1323	0.005	4×10^{-5}	3	PI3K_Growth
Esophagogastric	1422	0.78	0.068	0	Endocrine
Renal Cell	568	0.25	0.022	1	CellCycle
Small Bowel	129	0.27	2×10^{-3}	1	Immune
Small Cell Lung	375	0.69	0.021	1	PI3K_Growth

Chromatin Remodeling appears in 12.9%, consistent with its meta-regulatory role. Endocrine and Immune account for fewer than 5%. Among patients with two or more channels severed ($n = 31{,}156$), 94.5% have at least one cell-intrinsic channel.

The progression ladder is:

- **1 channel**: PI3K/Growth (35.6%) or Cell Cycle (30.0%) dominate. Chromatin Remodeling: 12.9%. Immune: 0.7%.
- **2 channels**: PI3K/Growth rises to 59.9%, Cell Cycle to 56.8%. Chromatin Remodeling enters at 29.1%. Tissue Architecture at 25.4%.
- **3 channels**: PI3K/Growth at 78.4%, Cell Cycle 69.6%, Chromatin Remodeling 52.1%, Tissue Architecture 50.0%. Endocrine (9.9%) and Immune (4.1%) remain rare.
- **4 channels**: PI3K/Growth at 87.7%, Chromatin Remodeling 75.7%. Endocrine and Immune are the last additions (16.4%, 9.9%).

The ordering is PI3K/Growth → Cell Cycle → Chromatin Remodeling → Tissue Architecture → DDR → Endocrine → Immune. Growth escape is the gateway; immune evasion is the last system to fall. Chromatin Remodeling appearing third is consistent with its Tier 3 (meta-regulatory) role: disrupting the machinery that controls gene expression enables subsequent channel breaks. Single-channel patients reflect first hits, not base rates. Section 3 interprets this as the escalation chain read in reverse. Figure 5 shows the conditional co-severance probability: given one channel is severed, how often is each other channel also severed. PI3K/Growth conditions nearly everything; Immune conditions almost nothing. The matrix is asymmetric: severing a cell-intrinsic channel predicts organism-level severance, but not the reverse.

The 5.5% of multi-channel patients who lack a cell-intrinsic gateway are enriched for prostate, breast, and mesothelioma: cancers entering through organism-level channels. Their lower mortality (37.8% vs. 42.3%) is consistent: they have not escaped the most fundamental coupling layer.

The grouping must be biological, not statistical. Hierarchical clustering on Jaccard co-mutation distances ($k = 2$ to $k = 10$) produces data-driven groupings that fail. At $k = 6$: HR = 0.80 ($p = 9 \times 10^{-6}$), the wrong direction. Biological channels at $k = 6$: HR = 1.10 ($p = 3 \times 10^{-9}$). ARI between data-driven and biological groupings: 0.018. Co-mutation clusters group genes by which tumors they appear in. Channels group genes by which organizational system they serve.

Pathogenic variant restriction. Restricting to truncating mutations only, then truncating plus hotspot missense, the signal strengthens: HR rises from 1.15 to 1.16 to 1.19 ($p = 10^{-55}$). Mutation count becomes more protective (HR = 0.75 vs. 0.83 baseline). Removing VUS concentrates the signal. The severance hierarchy is preserved.

Panel bias. MSK-IMPACT sequences 400–500 genes, a fraction of the genome. Mutations in channel genes not on the panel are undetected, systematically undercounting channel hits. This compresses the channel-count gradient and works against the hypothesis, not for it. The true association between channel count and survival is likely stronger than what the panel captures.

Limitations. Stage at diagnosis and treatment history are not available in these datasets. Channel count could correlate with stage. Three partial controls mitigate this: (1) within-cancer-type analysis, where stage distributions are more homogeneous; (2) the 275-covariate model includes FGA and tumor purity, which are stage-correlated proxies; (3) MetTropism is an all-metastatic cohort where stage variation is minimal, and the signal replicates. None of these fully substitute for stage adjustment. The analysis also does not distinguish hub from leaf mutations within a channel. The projection family predictions (Section 4) are untested.

3 Structural Interpretation: Genome as Projection

Mutations damage individual nodes. Channel damage destroys the distributed connectivity between nodes. This section explains why.

The projection forces three compromises. First, cycles must be broken. A linear sequence can only encode a DAG. Feedback loops are severed during projection and reconstructed at runtime via transcription factors, signaling cascades, and chromatin looping. These reconstructed edges are the most fragile.

Second, hub nodes must be copied. A single address cannot serve all adjacencies. The projection writes copies at different addresses. What appears as genetic diversity is serialization overhead.

Third, distributed connectivity cannot be written at all. Channels exist only in the three dimensional fold of the chromatin: unprecipitated knowledge, reconstructed by physical proximity when the helix folds.

Bounded context and escalation. A cell can maintain a finite number of regulatory states (C_n) [2]. Multicellular organization overcomes this through hierarchical coupling: cell ($k = 0$) escalates to tissue ($k = 1$), organ ($k = 2$), organism ($k = 3$) [7]. Coupling channels are the independent dimensions of this hierarchy: gap junctions at the cell-tissue boundary [1], paracrine signaling at tissue-organ, endocrine and immune at organ-organism [4].

Chain severance. When a channel is lost, the cell's context contracts. Actions requiring tissue level coordination become inaccessible [5, 6]. What remains: proliferation, metabolic autonomy, motility. Hanahan and Weinberg's hallmarks [4] are not acquired capabilities. They are the convergent action space under cell level bounded context. The Warburg effect is not a metabolic choice; it is the default when tissue level information about oxygen supply is gone [14].

Each channel severed removes an independent class of coupling. Mutation count measures depth within channels already severed; deeper commitment to fewer escape routes produces a more constrained tumor. The severance hierarchy (growth first, immune last) is the escalation chain read in reverse: cell-tissue coupling breaks before organism level coupling can be meaningfully exploited.

Two simultaneous channel breaks in one cell are lethal: the cell loses two independent defense axes. Channel count at the patient level therefore measures clonal diversity, not per-cell damage. Multi-region sequencing confirms this architecture [19, 20]: trunk mutations (the initiating channel break) are clonal; subsequent channel breaks appear as subclonal branch events, with different tumor regions independently converging on the same pathway through parallel evolution. The tumor does not acquire new channels by choice. The defense graph topology dictates which channel must break next; the specific gene is stochastic.

Projection families. Genes sharing interaction neighborhoods, mutation profiles, channel assignments, and survival impact are projections of the same node. The genome maintains redundant copies of critical subgraphs like a RAID array. Lose one copy and the system degrades. Remove the parity check (the synthetic lethal partner) and the degraded array fails.

Drug response follows projection families, not gene names. A PARP inhibitor works regardless of which DDR copy is broken: BRCA1, BRCA2, RAD51C, or any other projection of the same node. The drug targets function, not address.

Chain severance occurs through multiple mechanisms: genetic, microenvironmental, epigenetic, spatial (Table 5). All produce the same outcome, explaining why hereditary, environmental, and sporadic cancers converge on the same phenotypic endpoints.

4 Predictions

Five classes of prediction follow from the projection framework: expanded eligibility, combination futility, combination timing, untested targets, and immunotherapy sequencing.

4.1 DDR family: PARP inhibitor eligibility expansion

Current PARP inhibitor trials screen for a fixed list of DDR genes: *BRCA1*, *BRCA2*, *ATM*, *PALB2*, *RAD51C*, *RAD51D*, *FANCA*, *FANCC*, *FANCD2*. The framework predicts that additional DDR genes sharing the same protein-protein interaction neighborhoods, mutation type profiles, and survival impact should respond identically. Candidates include *BLM*, *NBN*, *ERCC4*, *CHEK1*, *CHEK2*, *RAD51B*, *RAD51*, *BRIP1*, *MRE11*, and *XRCC2*. In the MSK-IMPACT 44,000-patient cohort, these genes collectively account for over 4,000 patients—nearly 10% of the cohort—none of whom are currently eligible for PARP inhibitor trials.

Testable claim: Patients with loss-of-function mutations in *BLM*, *NBN*, *ERCC4*, *BRIP1*, or *MRE11* will show PARP inhibitor sensitivity comparable to *BRCA2* carriers, as measured by objective response rate and progression-free survival.

4.2 PI3K family: AKT/mTOR inhibitor eligibility expansion

PIK3CA and *PTEN* respond to PI3K/AKT/mTOR inhibitors. *PIK3C2G*, *INPP4B*, *PIK3CB*, *PIK3CD*, and *PIK3R3* are members of the same functional family with overlapping interaction neighborhoods but no drug sensitivity data.

Testable claim: Loss-of-function mutations in *INPP4B*, *PIK3CB*, and *PIK3CD* confer sensitivity to AKT inhibitors (e.g., capivasertib) comparable to *PTEN*-loss patients.

4.3 FGF family: FGFR inhibitor eligibility expansion

FGFR1, *FGFR2*, and *FGFR3* respond to FGFR inhibitors (erdafitinib, futibatinib). *FGFR4* and *FGF4* are functional family members with no targeted therapy data.

Testable claim: Activating mutations in *FGFR4* confer sensitivity to pan-FGFR inhibitors comparable to *FGFR2/3* fusion carriers.

4.4 Chromatin remodeling: CREBBP/EP300

CREBBP and *EP300* form one of the strongest functional pairs in the model (3,491 combined patients). Both are histone acetyltransferases with overlapping PPI neighborhoods and near-identical mutation profiles. Neither has established targeted therapy.

Testable claim: *CREBBP* and *EP300* mutations should respond to the same targeted therapy, once one exists. The projection framework predicts functional equivalence; it does not identify the drug. This is a prediction about family structure, not a treatment recommendation.

4.5 Same-channel combinations fail

The framework predicts that combining two drugs targeting the same coupling channel provides additive benefit at best, with compounding toxicity. The second drug adds dose, not a new therapeutic axis.

Testable claim: PARP + ATR inhibitors, PARP + WEE1 inhibitors, PARP + ATM inhibitors, and dual PARP1/2 versus PARP1-selective will show no significant PFS improvement over the simpler regimen after adjusting for toxicity. Dual targeting within a channel is architecturally redundant.

4.6 Cross-channel combination timing

Cross-channel combinations target two independent vulnerabilities. The framework predicts they work only when *both channels are intact at treatment*.

Metastatic futility: In metastatic HR+ breast cancer patients post-endocrine progression, adding a SERD (camizestrant) to a PARP inhibitor (saruparib) will show minimal additional PFS over PARP monotherapy. The endocrine escape has already occurred.

Adjuvant harmonization: In the adjuvant setting, low-dose PARP1-selective inhibitor alongside endocrine therapy will produce synergistic benefit in germline DDR carriers. The PARP kills cells at DDR second-hit before they accumulate ESR1 resistance mutations. The endocrine therapy maintains the channel those cells would have escaped through. Each drug extends the other's efficacy.

Tamoxifen works adjuvantly because it maintains the endocrine channel before severance. The same logic applies to DDR. PARP1-selective inhibitors (saruparib) eliminate PARP2-driven hematological suppression, enabling the long-term low-dose administration that adjuvant use requires. OlympiA showed adjuvant olaparib is tolerable alongside endocrine therapy; a PARP1-selective agent should be more tolerable still.

4.7 Testable observation: natural PARP1 disruption

If projection families share drug vulnerability, then naturally occurring *PARP1* loss-of-function mutations should produce the same synthetic lethality as PARP inhibitors in patients with DDR partner mutations.

Testable claim: Patients with *PARP1* loss-of-function alongside DDR partner mutations will show improved survival relative to DDR-mutant patients without *PARP1* mutations. A properly powered study with matched controls would constitute a strong test of the synthetic lethality mechanism independent of drug administration.

All drug-response predictions follow from one principle: screen by functional node, not by gene name.

4.8 Graph position predictions

If projection families define which genes share a vulnerability, graph position determines which phenotype results. Hub mutations (high-connectivity, upstream) should produce worse outcomes and broader phenotypes than leaf mutations (downstream, specific effectors) in the same channel, because hubs sever more downstream connections per event.

Testable claim: Hub mutations carry worse prognosis than leaf mutations within the same channel, after controlling for channel count and cancer type.

Testable claim: BRCA1 (DDR hub) and BRCA2 (DDR leaf) produce divergent breast cancer phenotypes. BRCA1 mutations enrich for triple-negative subtype and aggressive metastatic tropism. BRCA2 mutations enrich for HR+ subtype and bone-predominant metastasis. RAD51C, as a leaf-position DDR gene, phenocopies BRCA2, not BRCA1.

Companion paper [9] tests these graph-position predictions, including the distinction between gain-of-function (which maintains coupling) and loss-of-function (which severs it), and the directional escalation pattern within and across channels.

4.9 Channel severance beyond genetics

Channel severance occurs through any mechanism that disrupts the coupling between organizational levels (Table 5). Genetic mutation is one route. Immunosuppression, neurodegeneration, environmental disruption, and iatrogenic intervention are others. If the framework is correct, non-genetic channel severance should produce the same cancer risk predictions as genetic severance.

Testable claim: Organ transplant recipients on long-term immunosuppression have artificially severed immune channels. The framework predicts elevated cancer risk concentrated in immune-surveillance-dependent cancer types, not uniform elevation across all cancers. This is observed [10].

Non-genetic channel severance produces predictions spanning psychiatry, neurology, and epidemiology. Companion paper [11] develops these formally; here we note only that the framework's logic applies to any severance mechanism, not exclusively genetic mutation.

4.10 Immunotherapy candidacy after channel-level therapy

Channel-level therapies—hormone therapy, PARP inhibitors—suppress the channel they target but do not eliminate the tumor. Cancer cells under sustained suppression continue accumulating

mutations, but with the primary escape route blocked, the majority are passengers: they confer no selective advantage but each produces neoantigens visible to the immune system.

The framework predicts that effective channel-level therapy inadvertently marks residual cancer cells for immune recognition. The longer the therapy holds, the more passenger mutations accumulate, and the richer the neoantigen profile becomes. This produces two testable predictions:

Prediction 1: Passenger accumulation under therapy. Patients responding to channel-level therapy (adjuvant hormone therapy, PARP inhibitors) will show increasing passenger mutation burden over time, detectable via serial liquid biopsy (ctDNA). These passengers are not drivers of resistance—they are failed escape attempts. Each represents a neoantigen the immune system can target.

Prediction 2: Long-term tamoxifen patients who progress are immunotherapy candidates. ER+/PR+ breast cancer patients who respond to adjuvant tamoxifen for five or more years and then progress will respond to immune checkpoint inhibitors at rates exceeding current expectations for hormone-receptor-positive disease. The selection criterion is not receptor status or TMB at diagnosis; it is duration of prior adjuvant response. Years of suppressed-but-present disease produce a clean neoantigen profile—mostly passengers, few drivers—that is ideal for immune targeting.

ER+ breast cancer is considered non-immunogenic based on low PD-L1 expression and low TMB at diagnosis. The framework predicts this assessment is incorrect for the specific subpopulation of long-term adjuvant responders at first progression, whose neoantigen profile at the time of progression differs fundamentally from their profile at diagnosis.

Triple-negative breast cancer responds to checkpoint inhibitors because genomic instability produces abundant neoantigens rapidly. The long-term adjuvant hormone patient reaches a comparable neoantigen state through a different route: slow accumulation under therapeutic suppression. The marks are cleaner (fewer co-occurring driver mutations) but narrower (fewer total neoantigens). Whether this cleaner profile produces better or worse checkpoint response is an empirical question the framework identifies but cannot resolve without data.

Testable claim: Among ER+ breast cancer patients treated with checkpoint inhibitors at progression, objective response rate will correlate with duration of prior adjuvant endocrine therapy, not with TMB at initial diagnosis.

Prediction 3: TMB conditioned on channel count predicts immunotherapy response. TMB is a weak predictor of overall survival (Section 2) because it conflates two signals: channel escalations (driver mutations that breach new defense layers) and passengers (mutations that breach nothing). For survival, only escalations matter. For immunotherapy response, passengers matter —each is a neoantigen the immune system can target.

The framework predicts that TMB conditioned on channel count will outperform TMB alone as a predictor of checkpoint inhibitor response. A patient with high TMB and few disrupted channels carries mostly passengers—abundant neoantigens, few escape routes, ideal for immune targeting. A patient with the same TMB spread across many channels carries mostly escalations—each mutation opened a real escape route, leaving fewer marks and more resistance.

Triple-negative breast cancer illustrates this directly. TNBC tumors that respond to checkpoint inhibitors should have high TMB relative to their channel count—cancers that failed many escape attempts and left fingerprints. TNBC tumors that resist checkpoint should have low TMB relative to channel count—cancers that found escape routes efficiently, leaving few marks for immune to target.

Testable claim: Among checkpoint-treated patients in any cancer type, the ratio of mutation count to channel count will predict objective response rate. High ratio (many passengers per channel) predicts response; low ratio (few passengers per channel) predicts resistance. This can be tested immediately using existing clinical trial datasets that include both genomic sequencing and checkpoint inhibitor outcomes.

5 Engagement with Prior Work

Table 6 summarizes the relationship between existing cancer frameworks and chain severance.

The closest existing framework is TOFT [12]. It correctly identifies the organizational level (tissue, not cell) and the default proliferative state. Our framework provides the formalism TOFT has lacked: the escalation chain is the formal structure of tissue organization, C_n^{eff} is the measure of organizational integration, and chain severance is the formal description of organizational breakdown. TOFT's "default proliferative state" is our "necessary action space under cell level C_n."

No existing cancer model addresses the projection layer. Current frameworks treat genes as independent entities that participate in shared pathways. The projection view treats them as copies of shared functional nodes. This changes what it means to "target a gene."

6 Conclusion

Channel count predicts cancer survival ($p < 10^{-44}$). Mutation count does not.

The structural interpretation: the genome is a linear projection of a functional graph. Channels are the distributed connectivity that no single gene encodes. Genes in the same projection family carry the same drug vulnerability. Graph position within a family determines phenotype. Channel severance by any mechanism, genetic or otherwise, produces the same outcome.

This generates falsifiable predictions across four domains: drug eligibility expansion (PARP, PI3K/AKT, FGFR), combination therapy timing, graph position phenotyping, and non-genetic channel severance (transplant immunosuppression, neurodegeneration, information-environment cancer risk).

Screen by function, not by gene name. Companion papers test these predictions [9, 11]. Formal derivations appear in [5, 6, 7, 8].

Acknowledgments

The author used Claude (Anthropic) for drafting assistance, literature review, and LaTeX preparation. All theoretical content, proofs, and arguments are the author's own.

References

[1] T. Aasen, M. Mesnil, C. C. Naus, P. D. Lampe, and D. W. Laird. Gap junctions and cancer: Communicating for 50 years. *Nature Reviews Cancer*, 16(12):775–788, 2016.

[2] N. Cowan. The magical number 4 in short-term memory: A reconsideration of mental storage capacity. *Behavioral and Brain Sciences*, 24(1):87–114, 2001.

[3] P. C. W. Davies and C. H. Lineweaver. Cancer tumors as Metazoa 1.0: Tapping genes of ancient ancestors. *Physical Biology*, 8(1):015001, 2011.

[4] D. Hanahan and R. A. Weinberg. Hallmarks of cancer: The next generation. *Cell*, 144(5):646–674, 2011.

[5] P. D. McCarthy. Graph structure is necessary for information preservation under bounded context. Manuscript, 2026.

[6] P. D. McCarthy. Knowledge as normalization: consistency under partition separates shared propositions from action. Manuscript, 2026.

[7] P. D. McCarthy. Convergence of control architectures to escalation under bounded context. Manuscript, 2026.

[8] P. D. McCarthy. Evaluation-driven descent, encoding permanence, and the structural invariants of intelligence. Manuscript, 2026.

[9] P. D. McCarthy. Mutation position in the signaling graph predicts cancer survival beyond channel count. Manuscript, 2026.

[10] E. A. Engels, R. M. Pfeiffer, J. F. Fraumeni, B. L. Kasiske, A. K. Israni, J. J. Snyder, R. A. Wolfe, N. P. Goodrich, A. L. Bayakly, G. L. Clarke, and others. Spectrum of cancer risk among US solid organ transplant recipients. *JAMA*, 306(17):1891–1901, 2011.

[11] P. D. McCarthy. Context window diseases and cancer risk: Neural computation cost as a unifying mechanism. Manuscript, 2026.

[12] C. Sonnenschein and A. M. Soto. Somatic mutation theory of carcinogenesis: Why it should be dropped and replaced. *Molecular Carcinogenesis*, 29(4):205–211, 2000.

[13] B. Vogelstein, N. Papadopoulos, V. E. Velculescu, S. Zhou, L. A. Diaz, and K. W. Kinzler. Cancer genome landscapes. *Science*, 339(6127):1546–1558, 2013.

[14] O. Warburg. On the origin of cancer cells. *Science*, 123(3191):309–314, 1956.

[15] E. Cerami, J. Gao, U. Dogrusoz, B. E. Gross, S. O. Sumer, B. A. Aksoy, A. Jacobsen, C. J. Byrne, M. L. Heuer, E. Larsson, and others. The cBio cancer genomics portal: An open platform for exploring multidimensional cancer genomics data. *Cancer Discovery*, 2(5):401–404, 2012.

[16] A. Zehir, R. Benayed, R. H. Shah, A. Syed, S. Middha, H. R. Kim, P. Srinivasan, J. Gao, D. Chakravarty, S. M. Devlin, and others. Mutational landscape of metastatic cancer revealed from prospective clinical sequencing of 10,000 patients. *Nature Medicine*, 23(6):703–713, 2017.

[17] B. Nguyen, M. G. M. Fong, R. K. Luthra, S. M. Smith, R. J. DiNatale, N. Nandakumar, H. Walch, W. K. Chatila, R. Madupuri, R. Kundra, and others. Genomic characterization of metastatic patterns from prospective clinical sequencing of 25,000 patients. *Cell*, 185(3):563–575, 2022.

[18] C. Bandlamudi, J. J. Harding, A. Zehir, and others. Clinical genomic profiling of 50,000 patients: Experience from the MSK-IMPACT clinical sequencing cohort. *Cancer Cell*, 2026.

[19] M. Gerlinger, A. J. Rowan, S. Horswell, J. Larkin, D. Endesfelder, E. Gronroos, P. Martinez, N. Matthews, A. Stewart, P. Tarpey, and others. Intratumor heterogeneity and branched evolution revealed by multiregion sequencing. *New England Journal of Medicine*, 366(10):883–892, 2012.

[20] M. Jamal-Hanjani, G. A. Wilson, N. McGranahan, N. J. Birkbak, T. B. K. Watkins, S. Veeriah, S. Shafi, D. H. Johnson, R. Mitter, R. Rosenthal, and others. Tracking the evolution of non-small-cell lung cancer. *New England Journal of Medicine*, 376(22):2109–2121, 2017.

[21] C. Swanton, N. McGranahan, G. A. Wilson, and others (TRACERx Consortium). The evolution of lung cancer and impact of subclonal selection in TRACERx. *Nature*, 616(7958):525–533, 2023.

[22] P. C. Nowell. The clonal evolution of tumor cell populations. *Science*, 194(4260):23–28, 1976.

[23] S. Huang, I. Ernberg, and S. Kauffman. Cancer attractors: a systems view of tumors from a gene network dynamics and developmental perspective. *Seminars in Cell & Developmental Biology*, 20(7):869–876, 2009.

[24] C. A. Aktipis, A. M. Boddy, G. Jansen, U. Hibner, C. C. Maley, and G. S. Wilkinson. Cancer across the tree of life: Cooperation and cheating in multicellularity. *Philosophical Transactions of the Royal Society B*, 370(1673):20140219, 2015.

[25] L. M. F. Merlo, J. W. Pepper, B. J. Reid, and C. C. Maley. Cancer as an evolutionary and ecological process. *Nature Reviews Cancer*, 6(12):924–935, 2006.

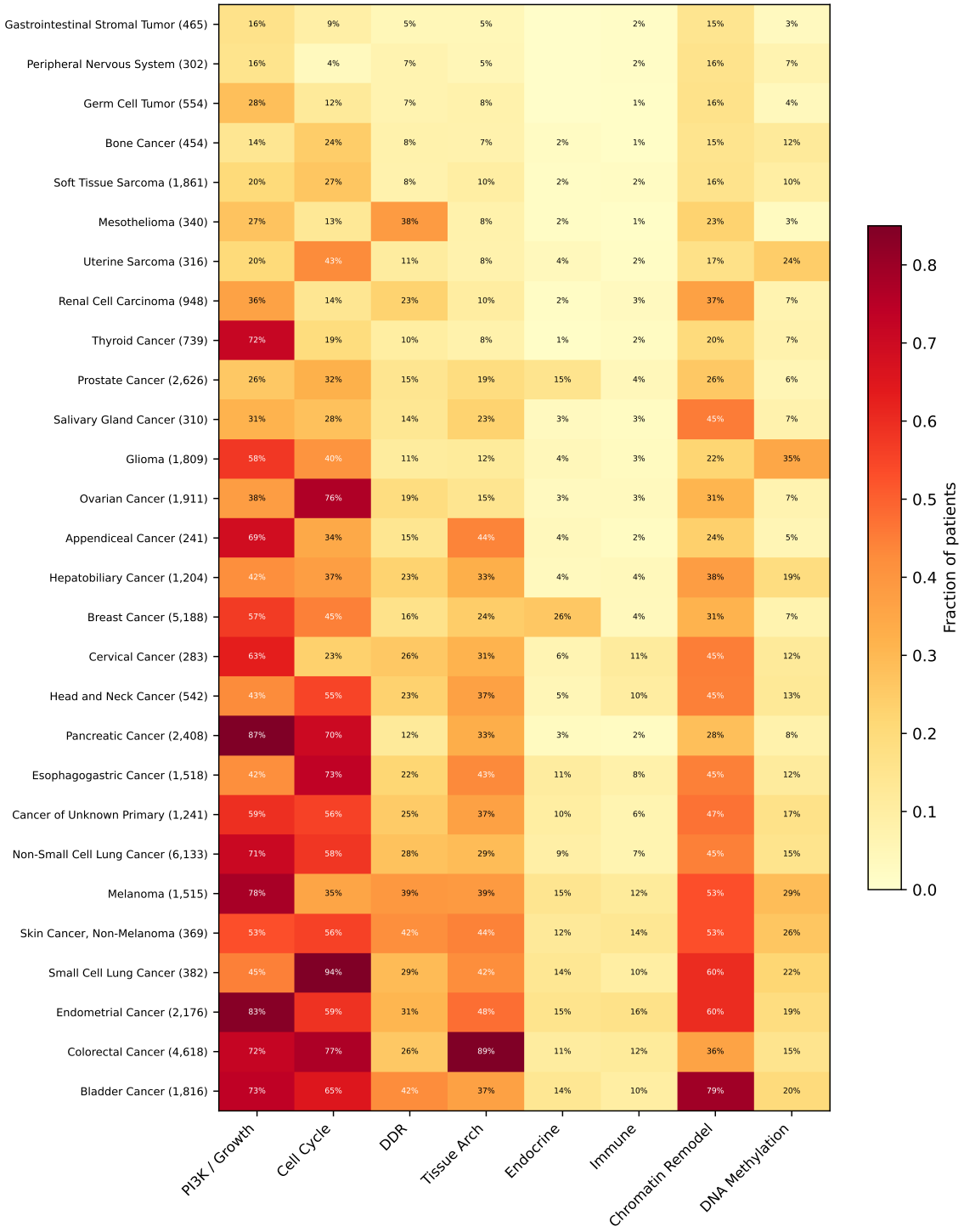

Figure 3: Channel severance frequency by cancer type. Each cell shows the fraction of patients in that cancer type with at least one mutation in the given channel. Cancer types are sorted by total channel burden. The tissue-specific pattern confirms that channels reflect organizational dependencies: breast is Endocrine-dominant, colorectal is Tissue Architecture-dominant, glioma is Cell Cycle-dominant.

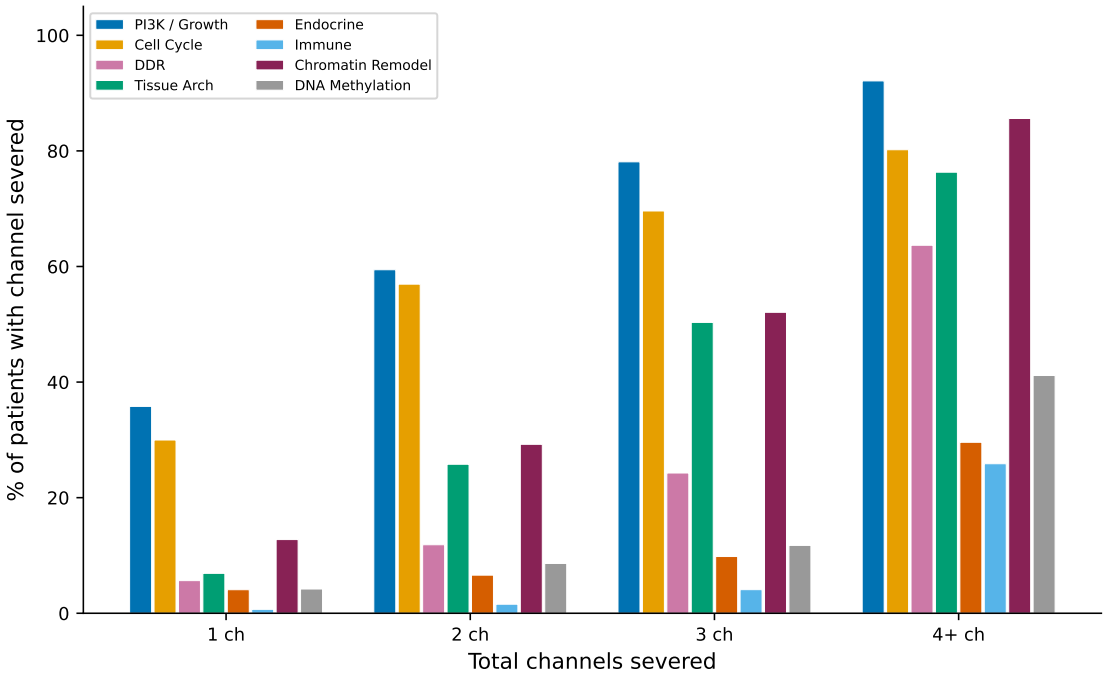

Figure 4: Severance hierarchy across channel-count strata (MSK-IMPACT 50K). Each bar shows the percentage of patients at a given stratum who have that channel severed. Cell-intrinsic channels (PI3K/Growth, Cell Cycle) dominate at every stratum; organism-level channels (Endocrine, Immune) appear only at higher channel counts.

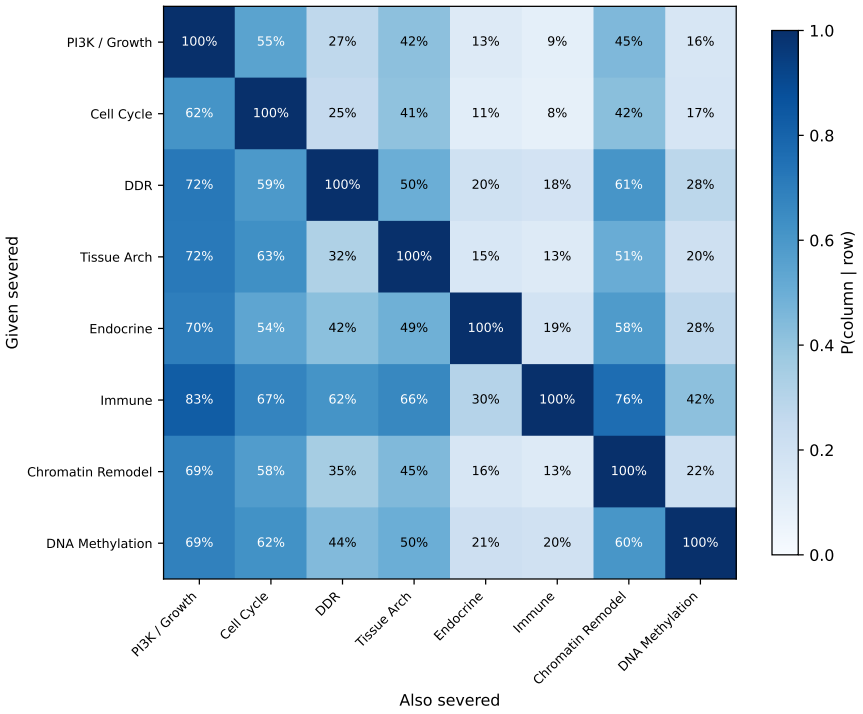

Figure 5: Channel co-severance matrix. Each cell shows P(column channel severed | row channel severed). The matrix is asymmetric: cell-intrinsic channels (PI3K/Growth, Cell Cycle) predict organism-level severance (Endocrine, Immune), but organism-level severance does not predict cell-intrinsic. Severance propagates inward toward the root of the escalation hierarchy.

Tables

Table 5: Mechanisms of chain severance. All four produce the same outcome: $\lambda_j \to 0$, context collapse to $C_n^{(0)}$, and cell-level optimization.

Mechanism	How $\lambda_j \to 0$	Example	Identified by
Genetic	Cell's receivers damaged	E-cadherin loss, p53/Rb mutation	SMT [13]
Microenvironmental	Signal no longer transmitted	Chronic inflammation, fibrosis, tissue damage	TOFT [12]
Epigenetic	Receiver genes silenced without sequence change	Methylation of connexin or cadherin promoters	—
Spatial	Cell leaves signaling field	Basement membrane breach, metastasis	—

Table 6: Relationship to existing cancer frameworks.

Framework	What it captures	Our reinterpretation
Hallmarks [4]	Universal capabilities across cancers	Not acquired but *revealed*: necessary action space under $C_n^{(0)}$
Atavistic theory [3]	Phylostratigraphic gene pattern	Convergent necessity, not historical reversion
Clonal evolution [22]	Darwinian selection dynamics	Provides the *attractor*: direction is determined by $C_n^{(0)}$, not random
Attractor landscape [23]	Cancer as abnormal attractors	"Abnormal" attractors are the *default* under cell-level C_n
Cancer as cheating [24]	Cooperation breakdown	Cheating is not a strategy but the necessary outcome of chain severance
Ecological process [25]	Tumor as ecosystem	Correct dynamics, but missing why they converge

Table 7: Complete gene-to-channel mapping used in all analyses. Genes are assigned based on organizational function served, not pathway membership alone. The full mapping (122 genes) and analysis code are provided in the supplementary repository.

Channel	**Genes**
DNA Damage Response	*ATM*, *ATR*, *BRCA1*, *BRCA2*, *PALB2*, *RAD51C*, *RAD51D*, *RAD51B*, *CHEK1*, *CHEK2*, *BAP1*, *BARD1*, *FANCA*, *FANCC*, *FANCD2*, *MLH1*, *MSH2*, *MSH6*, *PMS2*, *POLE*, *POLD1*
Cell Cycle / Apoptosis	*TP53*, *RB1*, *CDKN1A*, *CDKN1B*, *CDKN2A*, *CDKN2B*, *CDK4*, *CDK6*, *CCND1*, *CCNE1*, *MDM2*, *MDM4*, *MYC*, *MYCN*
PI3K / Growth Signaling	*PIK3CA*, *PIK3R1*, *PTEN*, *AKT1*, *AKT2*, *AKT3*, *MTOR*, *KRAS*, *NRAS*, *HRAS*, *BRAF*, *RAF1*, *MAP2K1*, *MAP2K2*, *MAP3K1*, *MAP3K13*, *ERBB2*, *ERBB3*, *EGFR*, *FGFR1*, *FGFR2*, *FGFR3*, *IGF1R*, *MET*, *NF1*, *NF2*, *TSC1*, *TSC2*, *STK11*, *ARID1A*
Endocrine / Hormone	*ESR1*, *ESR2*, *PGR*, *AR*, *FOXA1*, *GATA3*
Immune Surveillance	*B2M*, *HLA-A*, *HLA-B*, *HLA-C*, *JAK1*, *JAK2*, *STAT1*, *CD274*, *PDCD1LG2*, *CTLA4*
Tissue Architecture	*CDH1*, *CDH2*, *CTNNB1*, *APC*, *AXIN1*, *AXIN2*, *SMAD2*, *SMAD3*, *SMAD4*, *TGFBR1*, *TGFBR2*, *NOTCH1*, *NOTCH2*, *NOTCH3*, *NOTCH4*, *FBXW7*, *GJA1*, *GJB2*